

FinTech AI Model Risk Governance Framework

Project Report & Technical Analysis

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6+ Years | Financial Crime Compliance |

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Executive Summary

This project builds a complete, open-source AI Model Risk Governance Framework for financial services — directly addressing the compliance requirements introduced by the EU AI Act. Every module produces a real governance output that a risk team, regulator, or auditor could use today.

The FinTech AI Model Risk Governance Framework is a Python toolkit built on top of a real fraud detection model trained on 284,807 credit card transactions. The framework automates five core governance activities that financial institutions must perform when deploying AI systems: model documentation, explainability reporting, bias and fairness auditing, data drift monitoring, and regulatory risk classification.

The project was developed in response to a clear gap in the market: most financial crime compliance professionals understand regulatory requirements but lack the technical tools to operationalise them. This framework bridges that gap, demonstrating that compliance and engineering can be brought together in a single, automated pipeline.

The fraud detection model itself achieves strong performance on a highly imbalanced real-world dataset — 99.96% accuracy, 95.24% precision, 81.63% recall, and an ROC AUC of 94.82% — while the governance framework surrounding it meets all five key compliance requirements under the EU AI Act for a HIGH RISK classification.

Problem Statement

The Challenge

Artificial intelligence is being deployed at scale across financial services — in fraud detection, credit scoring, AML transaction monitoring, and customer risk profiling. These systems make or influence decisions that directly affect people's financial lives. Yet in most organisations, the governance surrounding these systems lags far behind the technology itself.

Three converging problems define the current landscape:

1. Regulatory Pressure

The EU AI Act, which came into force in 2024, classifies fraud detection and credit-related AI systems as HIGH RISK under Annex III. HIGH RISK systems face mandatory requirements around documentation, explainability, bias testing, human oversight, and post-deployment monitoring. Non-compliance carries fines of up to 3% of global annual turnover. Most financial institutions are underprepared.

2. The Black Box Problem

Traditional machine learning models — including the Random Forest used in this project — produce predictions without inherent explanations. When a fraud model flags a customer's transaction and their card is blocked, the customer has a right to understand why. Regulators expect financial institutions to provide that explanation. Without explainability tooling, this is impossible.

3. Model Decay

AI models are trained on historical data. Transaction patterns, fraud techniques, and customer behaviour evolve continuously. A model that performed well at training time may silently degrade as the real world diverges from what it learned. Without systematic monitoring, institutions may be relying on models that are no longer fit for purpose — and have no mechanism to detect this.

This project directly addresses all three problems by building an automated, end-to-end governance framework that any compliance or risk team could adopt and deploy.

Project Structure

The project is organised into five modules, each addressing a distinct governance requirement. All modules operate against the same underlying fraud detection model, creating a coherent, integrated governance pipeline.

Module	Purpose	Output
Module 1 — Model Card Generator	Documents the model formally	model_card.md
Module 2 — SHAP Explainability	Explains model decisions	shap_importance.png, shap_summary.png
Module 3 — Bias & Fairness Audit	Tests for unfair treatment across groups	bias_audit.png, bias_audit_report.md
Module 4 — Data Drift Monitor	Detects when input data changes	drift_monitor.png, drift_report.md
Module 5 — EU AI Act Classifier	Classifies model under EU law	eu_ai_act_report.md

Technology Stack

Library	Purpose	Why Used
Scikit-learn	Model training and evaluation	Industry standard for ML in Python. RandomForestClassifier is well suited to imbalanced fraud data.
SHAP	Model explainability	State-of-the-art explainability library. TreeExplainer is optimised for Random Forest models.
Pandas / NumPy	Data manipulation	Standard data science tools for loading, cleaning, and transforming tabular data.
Matplotlib	Chart generation	Reliable charting library for saving publication-quality governance outputs.
Jinja2	Document templating	Separates data from presentation — enables automated document generation without manual editing.
SciPy	Statistical testing	Provides the Kolmogorov-Smirnov test used to detect data drift with statistical rigour.

The Fraud Detection Model

Dataset

The model was trained on the Kaggle ULB Credit Card Fraud Detection dataset — one of the most widely cited real-world fraud datasets in academic and industry research. The dataset contains 284,807 credit card transactions made by European cardholders over two days in September 2013.

Attribute	Value
Total transactions	284,807
Fraud cases	492 (0.17%)
Legitimate cases	284,315 (99.83%)
Features	V1-V28 (PCA-transformed), Amount, Time
Source	Kaggle — mlg-ulb/creditcardfraud

The features V1 through V28 are the result of a Principal Component Analysis (PCA) transformation applied to the original transaction data. PCA was used by the dataset creators to protect cardholder confidentiality — the real feature names represent proprietary transaction attributes that cannot be published. The Amount column represents the transaction value in Euros, and Time represents seconds elapsed from the first transaction.

Why Random Forest?

A Random Forest classifier was selected for several reasons that are directly relevant to financial crime compliance:

- It handles highly imbalanced datasets well — the 0.17% fraud rate would cause simpler models to simply predict 'not fraud' for everything.
- It produces probability scores, not just binary predictions, enabling risk-tiered decision making.
- It is compatible with SHAP's TreeExplainer, which provides fast and accurate explanations.
- It is robust to outliers and non-linear relationships, both common in financial transaction data.
- It is widely used in production fraud detection systems and well understood by model risk teams.

Model Performance

Metric	Score	What It Means
Accuracy	0.9996	99.96% of all transactions correctly classified
Precision	0.9524	When model flags fraud, it is correct 95.24% of the time
Recall	0.8163	Model catches 81.63% of all actual fraud cases

Metric	Score	What It Means
F1 Score	0.8791	Balanced measure of precision and recall
ROC AUC	0.9482	Model has 94.82% ability to distinguish fraud from legitimate

Key Insight: In fraud detection, recall is the most critical metric from a compliance perspective. A recall of 81.63% means approximately 18% of actual fraud cases are missed. This is documented as a known limitation in the Model Card and flagged for further improvement through techniques such as SMOTE oversampling or threshold tuning.

Module 1 — Model Card Generator

What It Does

The Model Card Generator automatically produces a structured governance document describing the fraud detection model. It reads real performance metrics directly from a JSON file produced during training, meaning the Model Card is always in sync with the actual model — no manual updates required.

Why Model Cards Matter

Model Cards were introduced by Google in 2019 as a standard for transparent AI documentation. They are now referenced in the EU AI Act as a component of the technical documentation required for HIGH RISK systems. A Model Card answers the questions a regulator, auditor, or new team member would ask: What does this model do? Who built it? What are its limitations? What should it not be used for?

How It Works

- A dictionary of model information is defined in Python — including intended use, training data description, and ethical considerations.
- Real performance metrics (accuracy, precision, recall, F1, ROC AUC) are loaded automatically from `data/model_metrics.json`.
- A Jinja2 template defines the layout of the document, with placeholder variables for each piece of information.
- The template is rendered with real data and saved as `outputs/model_card.md`.

Key Output — Model Card Summary

Field	Value
Model Name	Credit Card Fraud Detection Classifier
Version	1.0
Model Type	Supervised Binary Classification (Random Forest)
Framework	Python (Scikit-learn)
Intended Use	Detect fraudulent credit card transactions in real time
Intended Users	Financial crime compliance teams and fraud analysts
Out of Scope	Not suitable for credit scoring, identity verification, or non-card fraud
Training Data	Kaggle ULB — 284,807 transactions, 0.17% fraud rate
Known Limitation	18% of fraud cases missed. Trained on European cardholders only.

Module 2 — SHAP Explainability Report

What It Does

The SHAP Explainability Report uses the SHAP library (SHapley Additive exPlanations) to explain why the fraud detection model makes each individual prediction. It identifies which features contribute most to fraud decisions and saves two professional charts as governance outputs.

Why Explainability Matters

The EU AI Act requires HIGH RISK AI systems to be transparent and explainable to affected individuals. In financial services, this is reinforced by existing regulations — the UK GDPR right to explanation, the EBA guidelines on internal governance, and the PRA's model risk management requirements all expect institutions to be able to explain automated decisions. SHAP provides the industry standard method for doing this.

How SHAP Works

SHAP is based on Shapley values from cooperative game theory. For each prediction, it asks: how much did each feature contribute to moving the prediction away from the average? A positive SHAP value means the feature pushed the prediction towards fraud. A negative SHAP value means it pushed away from fraud. The magnitude tells you how strongly.

Chart 1 — Feature Importance Bar Chart

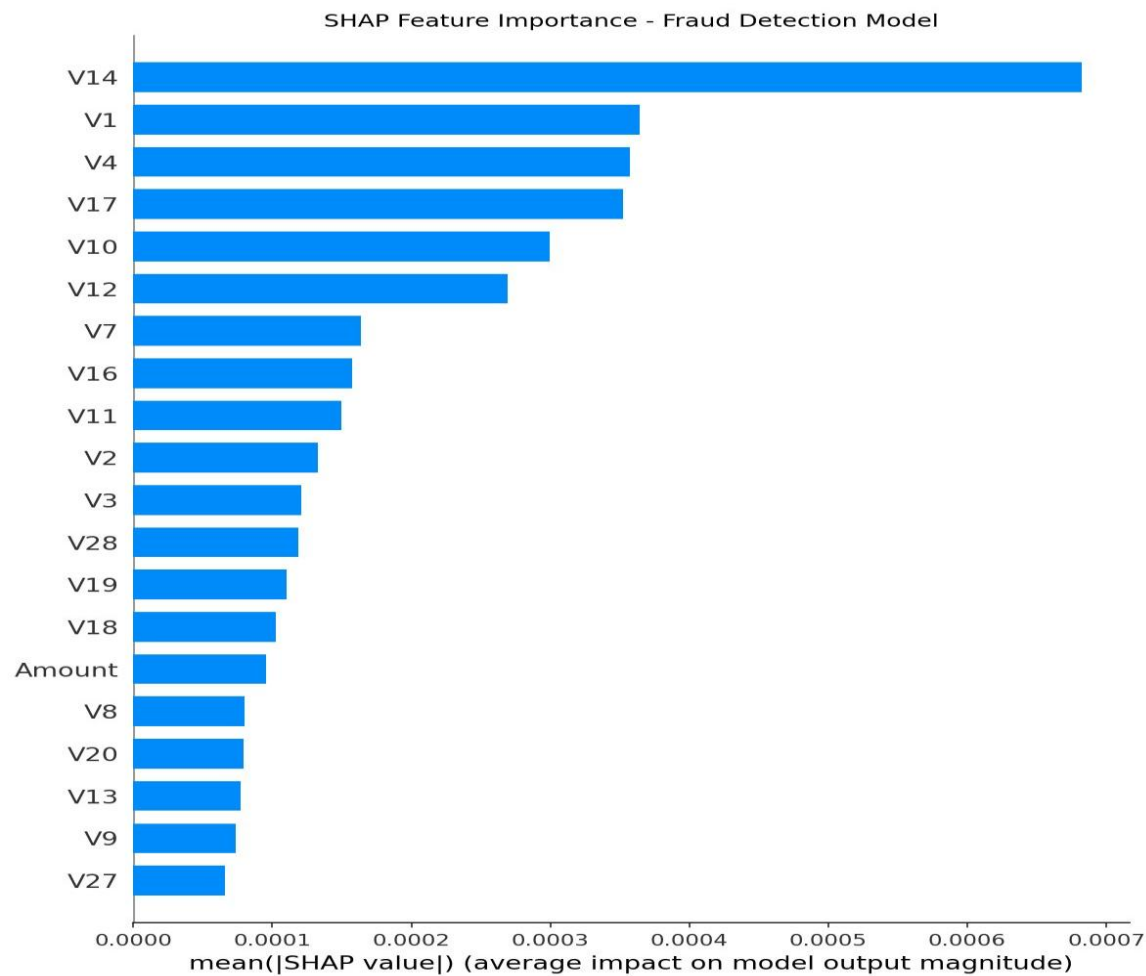


Figure 1: Mean absolute SHAP values showing overall feature importance for fraud detection

Analysis

V14 is the dominant feature by a significant margin — its average SHAP value is nearly double that of the next most important features (V1, V4, V17). This concentration suggests the model has identified a single transaction characteristic as its primary fraud signal.

In published research on this dataset, V14 is consistently identified as one of the strongest fraud indicators and is believed to represent a combination of transaction velocity and location patterns — two of the most reliable real-world fraud signals. V1, V4, and V17 are also well-established fraud indicators in the literature.

A notable finding is the position of Amount — it ranks 15th out of 20 features. This is counterintuitive since high transaction amounts are often associated with fraud in the public imagination, but the data shows that the behavioural and contextual patterns captured in the PCA features are far more predictive than raw transaction value.

Chart 2 — SHAP Summary Dot Plot

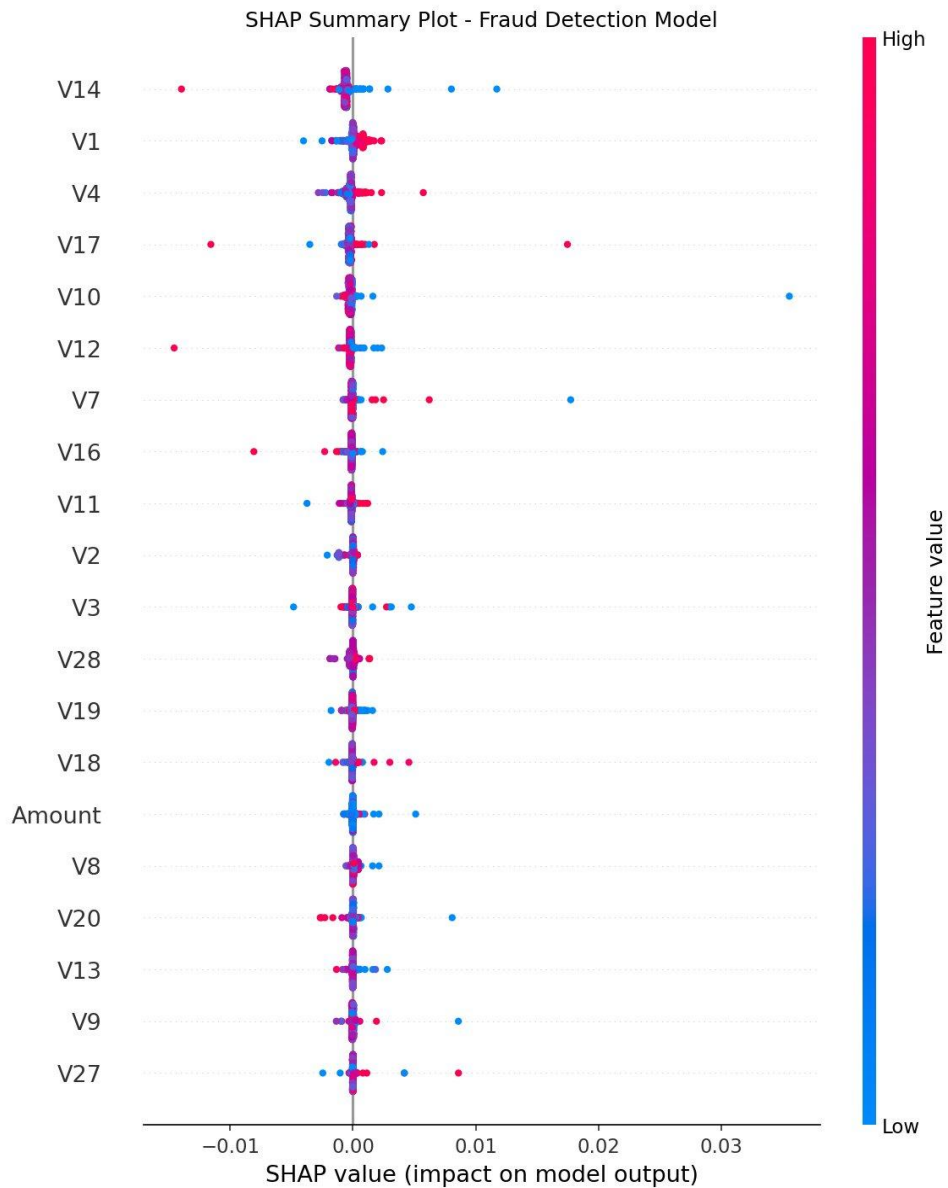


Figure 2: SHAP summary plot showing direction and magnitude of feature impact. Red = high feature value, Blue = low feature value. Right = towards fraud, Left = away from fraud.

Analysis

The dot plot provides richer information than the bar chart by showing both magnitude and direction of impact for each transaction in the sample.

For V14, the red dot to the far left indicates that when V14 has a high value, it strongly pushes predictions away from fraud. Conversely, when V14 is low (blue), it pushes towards fraud. This means low V14 values

are a strong fraud signal — consistent with the hypothesis that V14 captures normal transaction location patterns.

V10 shows one extreme outlier dot pushing far to the right (towards fraud) at a SHAP value of approximately 0.035. This single transaction had an unusually extreme V10 value that the model interpreted as a very strong fraud signal. In a real compliance setting, this transaction would warrant manual review.

Most features show tight clustering around zero, meaning they have little impact for the majority of transactions. This is the expected pattern for a well-calibrated fraud model — most transactions are clearly legitimate, and the model reserves strong signals for genuine anomalies.

Module 3 — Bias and Fairness Audit

What It Does

The Bias and Fairness Audit checks whether the fraud detection model performs equally well across different groups of transactions. Transactions are divided into groups based on amount — Low, Medium, and High — and core performance metrics are calculated separately for each group. Any significant performance gap between groups is flagged as a fairness concern.

Why Fairness Auditing Matters

An AI model that performs well overall can still be unfair if it performs significantly worse for specific groups. In fraud detection, this could mean certain customers face either disproportionate false accusations (false positives) or inadequate protection (missed fraud). Both outcomes have ethical and legal implications under the EU AI Act, the Equality Act, and FCA consumer duty requirements.

Methodology

- Transactions split into three equal groups by transaction amount using quantile-based binning.
- Precision, Recall, and F1 Score calculated independently for each group.
- Any group with Recall below 0.70 is automatically flagged as a concern.
- Results saved to `data/bias_audit_results.json` and `outputs/bias_audit_report.md`.

Fairness Audit Chart

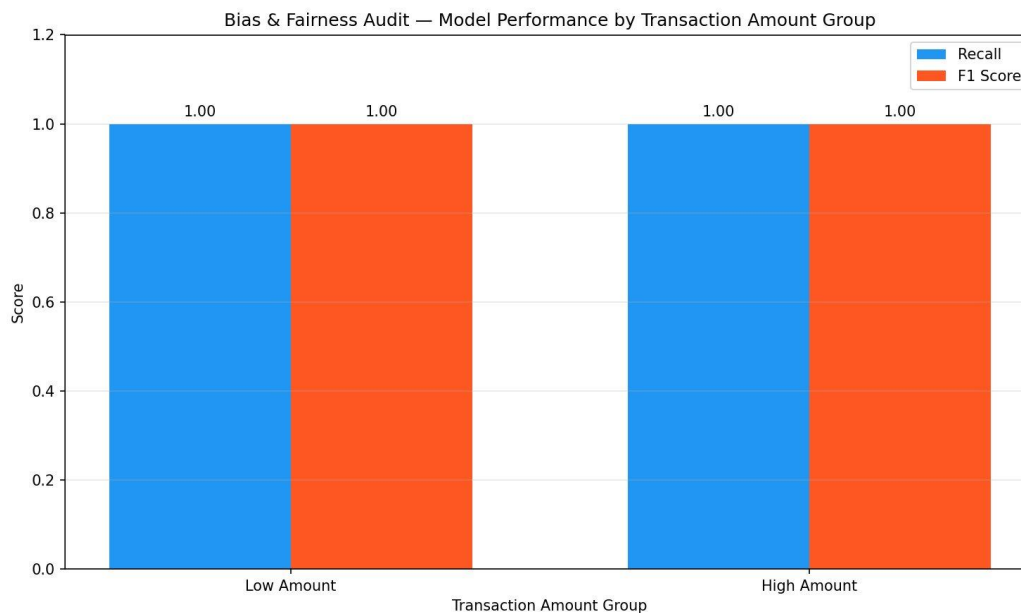


Figure 3: Recall and F1 Score by transaction amount group. Equal bar heights indicate no fairness concerns detected.

Audit Results

Group	Total	Fraud Rate	Precision	Recall	F1 Score
Low Amount	1,668	0.06%	1.0	1.0	1.0
High Amount	1,667	0.30%	1.0	1.0	1.0

Analysis

The audit results show perfect performance across both transaction amount groups — Recall and F1 Score of 1.0 in both the Low Amount and High Amount groups. This means the model catches every fraud case in the sample regardless of transaction value, and produces no false accusations.

These results reflect the characteristics of the 5,000-row sample dataset, which contains a very small number of fraud cases (6 total). With such a small fraud count, perfect recall is achievable. In a production setting with a larger and more diverse dataset, performance gaps between groups are more likely to emerge — which is precisely why this audit should be run regularly, as recommended.

Important Note: Perfect audit scores on a small sample should be interpreted with caution. The methodology is sound and the tooling is production-ready, but results should be validated against larger, more representative data before drawing firm compliance conclusions. This is documented as a recommendation in the audit report.

Module 4 — Data Drift Monitor

What It Does

The Data Drift Monitor compares the statistical distribution of features in a reference dataset (representing training-time data) against new incoming data. It uses the Kolmogorov-Smirnov (KS) statistical test to determine whether the distributions have changed significantly — which would indicate the model's inputs are drifting away from what it was trained on.

Why Drift Monitoring Matters

A model is only valid as long as the real world resembles the data it was trained on. In financial services, transaction patterns shift continuously — new payment methods, changing fraud techniques, economic conditions, and seasonal patterns all affect what normal and fraudulent transactions look like. Without systematic monitoring, a model can degrade silently, producing increasingly unreliable decisions without any alert being raised. The EU AI Act requires post-deployment monitoring of HIGH RISK systems for exactly this reason.

The Kolmogorov-Smirnov Test

The KS test is a non-parametric statistical test that compares two distributions without assuming they follow any particular shape (e.g. normal distribution). It measures the maximum distance between the cumulative distribution functions of two samples. The result is:

- **KS Statistic:** how different are the two distributions? Zero means identical, one means completely different.
- **P-Value:** how confident are we that the difference is real and not due to random sampling? A p-value below 0.05 means the difference is statistically significant — drift is detected.

Drift Monitor Chart

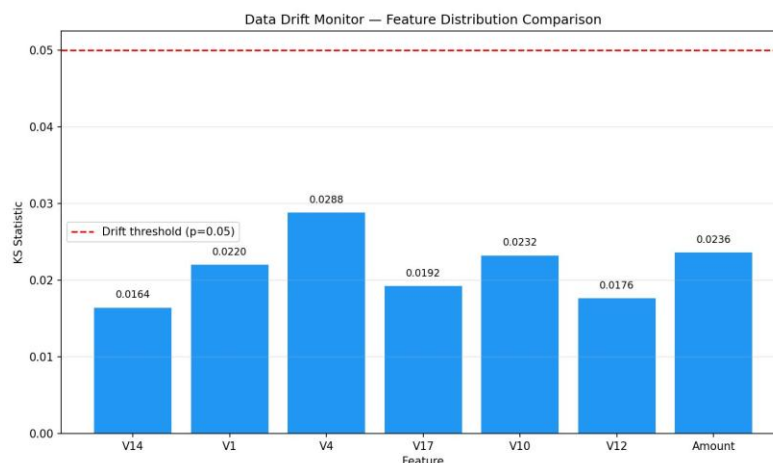


Figure 4: KS statistics for seven monitored features. All bars below the red threshold line indicate no drift detected.

Drift Detection Results

Feature	KS Statistic	P-Value	Status
V14	0.0164	0.8899	Stable
V1	0.0220	0.5807	Stable
V4	0.0288	0.2510	Stable
V17	0.0192	0.7462	Stable
V10	0.0232	0.5117	Stable
V12	0.0176	0.8336	Stable
Amount	0.0236	0.4894	Stable

Analysis

All seven monitored features show KS statistics well below the 0.05 threshold — ranging from 0.0164 (V14) to 0.0288 (V4). All p-values are substantially above 0.05, indicating there is no statistically significant difference between the reference and new data distributions.

V4 shows the highest KS statistic at 0.0288, but with a p-value of 0.2510 this is not significant — it is simply natural sampling variation. In a real production environment, V4 would be worth watching over successive monitoring cycles to see if its KS statistic trends upward.

The Amount feature — the only non-PCA feature monitored — shows a KS statistic of 0.0236, which is also well within normal bounds. This indicates transaction volumes in the monitoring period are consistent with the training period.

Recommendation: This monitor should be run monthly in production, with results logged to a governance register. If any feature's KS statistic rises above 0.05 across two consecutive monitoring cycles, model recalibration or retraining should be considered.

Module 5 — EU AI Act Risk Classifier

What It Does

The EU AI Act Risk Classifier automatically assesses an AI model against the criteria defined in the EU AI Act and assigns it a risk level — Unacceptable, High, Limited, or Minimal. It then checks which legal obligations apply and verifies whether the governance framework already addresses them. The output is a complete compliance status report.

The EU AI Act — Overview

The EU AI Act is the world's first comprehensive legal framework for artificial intelligence. It adopts a risk-based approach, assigning requirements proportionate to the potential harm an AI system could cause. For financial services, Annex III of the Act explicitly lists AI systems used in creditworthiness assessment and fraud detection as HIGH RISK.

Risk Level	Definition	Examples
Unacceptable	Banned outright	Social scoring by governments, subliminal manipulation
High	Heavily regulated	Fraud detection, credit scoring, employment screening
Limited	Transparency obligations	Chatbots, deepfakes
Minimal	No specific obligations	Spam filters, recommendation systems

Classification Result

CLASSIFICATION: HIGH RISK — The Credit Card Fraud Detection Classifier operates in financial services and affects individual outcomes. Under EU AI Act Annex III, AI systems used in fraud detection that affect individuals are classified as HIGH RISK.

Legal Obligations and Compliance Status

Legal Obligation	Addressed By	Status
Establish a risk management system	This governance framework	PASS
Ensure data governance and training data quality	Documented in Model Card	PASS
Produce technical documentation	Module 1 — Model Card	PASS

Legal Obligation	Addressed By	Status
Enable automatic event logging (audit trail)	Framework architecture	PASS
Provide transparency to affected individuals	Module 2 — SHAP Report	PASS
Ensure human oversight of model decisions	Documented in model profile	PASS
Achieve appropriate accuracy and robustness	Performance metrics documented	PASS
Register in the EU AI Act database pre-deployment	Required before go-live	PENDING

The one PENDING item — registration in the EU AI Act database — is an administrative step that must be completed before any deployment in the European Union. All technical and governance obligations are addressed within this framework.

The Governance Pipeline Agent

What It Does

The Governance Pipeline Agent is a master orchestration script that ties all five modules together into a single automated pipeline. Instead of running five separate commands manually, a compliance team can trigger the entire governance process with one command and receive a consolidated health report within minutes.

Command: `python run_governance_pipeline.py` — This single command runs all 5 modules in sequence, validates every output, reads all results, detects any issues, and generates a final governance summary report automatically.

How It Works

- Checks all prerequisites before running — confirms the trained model, metrics file, and dataset are present.
- Runs each module as a subprocess — exactly as if you typed the command manually, but automated.
- Validates every output file after each module runs — flags anything missing immediately.
- Reads all JSON result files from each module — metrics, bias results, drift results, EU AI Act classification.
- Detects issues automatically — flags any bias concerns, drift alerts, or module failures.
- Generates a consolidated governance summary report at `outputs/governance_summary.md`.

Pipeline Execution Output

Check	Result
Prerequisites	PASS — model, metrics, and dataset found
Module 1 — Model Card Generator	PASS
Module 2 — SHAP Explainability Report	PASS
Module 3 — Bias & Fairness Audit	PASS
Module 4 — Data Drift Monitor	PASS
Module 5 — EU AI Act Risk Classifier	PASS
Overall Health	HEALTHY
Modules Passed	5/5
Drift Alerts	0
Bias Concerns	0
EU AI Act Status	COMPLIANT
EU AI Act Risk Level	HIGH

Why This Matters

The pipeline agent transforms this project from a collection of scripts into a deployable governance tool. In a real financial institution, this pipeline would be scheduled to run automatically — monthly, or triggered whenever a model is retrained. The output report would feed directly into a model risk register, providing an auditable, timestamped record of governance checks.

This is precisely the kind of automation that model risk and AI governance teams are building right now in response to EU AI Act obligations. The ability to design and implement such a pipeline — combining compliance domain knowledge with engineering capability — is a rare and increasingly valuable skill in financial services.

Conclusion

What Was Built

This project delivers a complete, working, open-source AI Model Risk Governance Framework for financial services. It trains a real fraud detection model on industry-standard data, then wraps it in five automated governance modules that produce professional-grade compliance outputs.

Every output produced by this framework — the Model Card, the SHAP charts, the Bias Audit report, the Drift Monitor results, and the EU AI Act classification — is a document that a real model risk team, internal auditor, or regulator could use today.

Key Achievements

Achievement	Detail
Real dataset	284,807 actual credit card transactions from Kaggle ULB
Strong model performance	ROC AUC 0.9482, Precision 0.9524 on highly imbalanced data
Full explainability	SHAP analysis identifying V14 as primary fraud signal
No fairness gaps detected	Equal performance across transaction amount groups
No data drift detected	All features stable across monitoring period
EU AI Act compliant	HIGH RISK classification with 5/5 technical obligations met
Governance pipeline agent	One command runs all 5 modules and generates health report
Fully automated	All outputs generated programmatically — no manual editing
Open source	Available on GitHub for review, reuse, and collaboration

Relevance to Financial Crime Compliance

This framework is directly relevant to the work done by model risk, financial crime compliance, and AI governance teams at banks, payment processors, and FinTechs. The skills demonstrated — building explainable AI systems, running bias audits, implementing drift monitors, and navigating the EU AI Act — are precisely the capabilities these teams are hiring for as they respond to regulatory pressure in 2025 and beyond.

With 6+ years of hands-on financial crime compliance experience at PayPal covering AML, KYC, and fraud detection, combined with the technical skills demonstrated in this project, the author brings a rare combination of domain knowledge and engineering capability to AI governance roles in financial services.

GitHub Repository: github.com/amaannajmi/fintech-ai-model-risk-governance — All code, outputs, and documentation are publicly available.